

A Geometry-Material Aware Point Cloud Transformer for Large-scale Unstructured Thermal Analysis in 2.5D ICs

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Abstract—Thermal management in large-scale unstructured 2.5D ICs faces the challenges due to the integration of complex geometries and heterogeneous materials. Existing deep learning (DL) methods urgently require a memory-efficient and high-fidelity unstructured representation method for multiscale complex ICs to simultaneously model macroscopic components and microscopic structure. Moreover, it further needs to achieve multiscale geometric thermal feature capture and thermal distribution difference adaptation among heterogeneous materials. Combining a multiscale unstructured point-cloud representation, this paper introduces Therm-PCT, a geometry-material aware point-cloud transformer framework to achieve high-accuracy thermal and its gradient prediction. Therm-PCT incorporates three key modules: adaptive multipath-coupled diffusion (AMD), a wavelet-based fine-grained recovery (WFR), and a thermal-aware Mixture-of-Material-Experts (TA-MoME) adapter. AMD adaptively learns heat diffusion path interaction with serialization-gate-based attention. Furthermore, the WFR module recovers fine-grained thermal gradients through high-frequency wavelet domain enhancement, and the TA-MoME adapter adapts to heterogeneous material by dynamically routing material-specific experts. Experiments demonstrate that the Thermal-PCT’s accuracy performance metric improvements are substantial, outperforming the newly proposed method FSA-Heat, by considerable margins of 78.03%, 84.00%, 67.61%, and 78.25% in 80 K-scale point clouds. It also achieves a 147× speed-up compared to the commercial software COMSOL. Additionally, Therm-PCT shows the potential of zero-shot generalization up to 0.4 M-scale points (5.7× than training scale) and robust performance on unseen geometric shapes.

Index Terms—Thermal analysis, 2.5D ICs, Point cloud, Deep learning, Transformer.

I. INTRODUCTION

Under Moore’s Law, the exponential increase in device density and the ongoing scaling of transistors have led the semiconductor industry to adopt advanced packaging techniques like chiplet-based 2.5D/3D integrated circuit (IC) heterogeneous integration [1], [2]. While chiplet-based technology enhances performance and interconnection density, it introduces unprecedented thermal management challenges due to complex structural stacking, heterogeneous material, and spatially irregular power maps [3], [4]. The increasingly severe thermal gradient regions and hotspots will critically threaten device reliability and performance [5], [6]. Therefore, it necessitates accurate and

fast thermal analysis, which will be performed several times in the early-stage design.

Traditional numerical methods such as finite element methods (FEM) and finite difference methods (FDM) have long served as cornerstones for thermal simulation [6], [7]. Commercial solvers such as Celsius, ANSYS, and COMSOL widely employ the FEM for thermal simulation analysis [8], [9], [10], which provides the best accuracy and flexibility. However, these methods face significant computational costs when simulating modern IC designs with large-scale unstructured geometries, particularly in iterative thermal optimization workflows.

Recent advancements in deep learning (DL) indicate significant potential in accelerating 2.5D IC thermal prediction. Newly-proposed DL-based approaches, including deep operator networks [11], [12], convolutional neural networks (CNNs) [13], [14], [15], and graph neural network (GNN) [16], achieve thermal prediction instead of explicit PDE discretization via different operator methods: DeepOHeat leverages fixed basis function operator to capture thermal field mappings. CNN-based methods like FSA-Heat efficiently extract grid-structured thermal map features. GNN-based networks like the GCN+PNA can model unstructured geometries via node-edge topologies. However, these methods face distinct limitations when applied to large-scale, unstructured 2.5D IC thermal predictions.

Large-scale unstructured 2.5D ICs require simultaneous modeling of macroscopic components (e.g., substrate) and microscopic interconnect structures (e.g., bumps) [17]. While CNNs fail to effectively represent non-grid geometries due to rigid grid-structure convolutional layers, GNN faces scalability bottlenecks due to prohibitive memory overhead in full-system simulations [16]. Thus, *how to propose a memory-efficient and high-fidelity unstructured representation method for large-scale unstructured complex ICs is vital and under exploration.*

Moreover, compared to simple geometric representation, unstructured multiscale representations further introduce the two key thermal challenges: (1) **interfacial thermal discontinuities** caused by conductivity mismatches among heterogeneous materials [18]; and (2) **multiscale geometric thermal gradients** ranging from micron-level hotspots to full-chip profiles [19]. These complex physical challenges conflict with the exist-

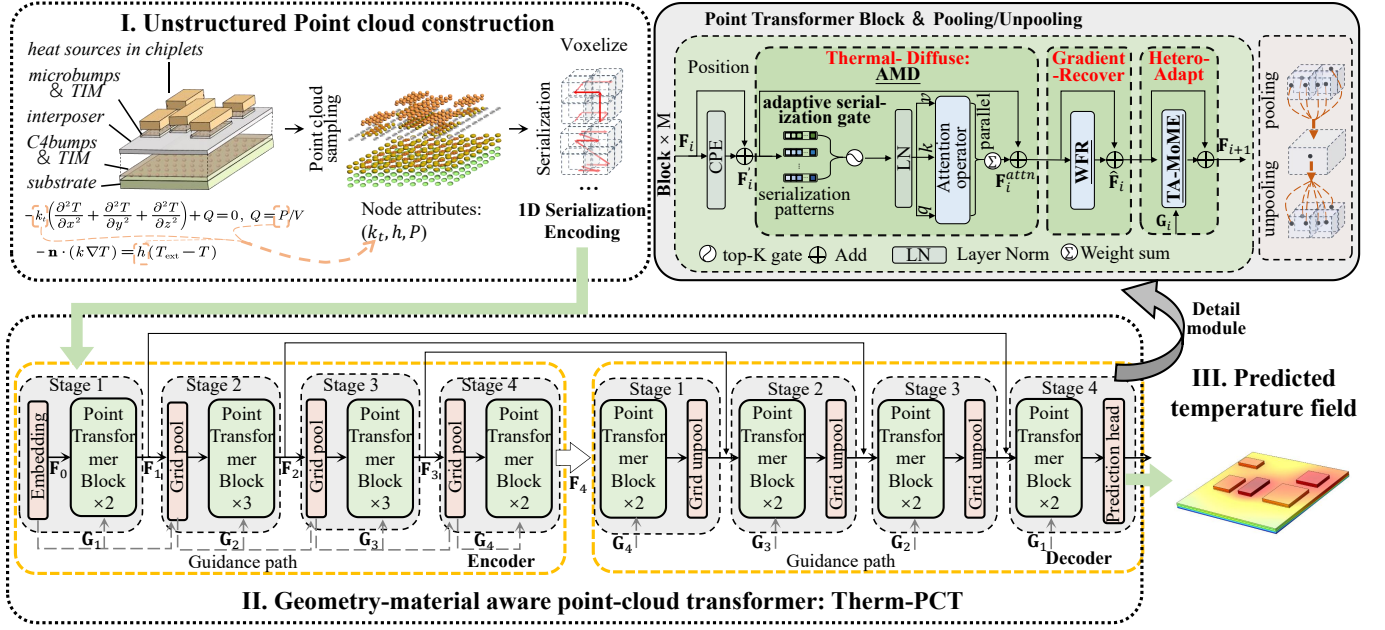


Fig. 1: The pipeline of point cloud processing (I.), Overall architecture of geometry-material aware point-cloud transformer (II.), and its final output (III.).

ing DL architectures [20], [21]. CNNs fail to resolve abrupt interfacial discontinuities due to fixed homogeneous kernels; DeepONet and GNN struggle to capture the thermal difference among heterogeneous materials (challenge 1) and fine-grained gradient details (challenge 2) due to intrinsic architectural constraints: GNNs exhibit oversmoothing thermal tendencies through graph aggregation, and DeepONet lacks explicit mechanisms for resolving multiscale feature interactions in different materials. Thus, *how to adapt to the thermal differences among heterogeneous materials and capture fine-grained geometric thermal features remains an open challenge.*

In this work, a geometry-material aware point-cloud transformer framework Therm-PCT is proposed, which firstly proposes a memory-efficient and high-fidelity point-cloud representation for 2.5D unstructured ICs. Furthermore, Therm-PCT jointly captures fine-grained thermal gradient features and differential material thermal distributions through the proposed fine-grained gradient recovery method and heterogeneous material adaptation. The main contributions are as follows:

- To the best of our knowledge, our work is the first to introduce a point-cloud representation architecture for large-scale unstructured 2.5D IC thermal prediction. We propose adaptive multipath-coupled diffusion (AMD) to dynamically capture thermal diffusion interaction across multiple 1D serialization pathways. The framework enables high-precision thermal and its gradient prediction for 80 K points, with zero-shot generalization potential for up to 0.4 M-scale points and unseen geometries.
- To recover the gradient feature details lost in micro-geometric structures (particularly bumps), caused by spectral smoothing in point cloud attention, a wavelet-based fine-grained recovery (WFR) module is designed. This module refines fine-grained thermal features by enhancing

high-frequency subband features in the decoupled wavelet domain. It enhances gradient granularity perception in multiscale geometries.

- To adapt to the large temperature distribution difference among materials, a thermal-aware Mixture-of-Material-Experts (TA-MoME) adapter is proposed. A material-expert router is designed to consider both heterogeneous physical parameters and fine-grained location features. Specific experts are activated to adapt to heterogeneous materials exhibiting varying thermal distributions.

II. PROBLEM SETTING AND PRELIMINARIES

A. Heat conduction problem

In 2.5D integrated circuits, the heat conduction process can be described by the classical Fourier's Law [22]. The steady-state heat conduction problem within the interior region is governed by the following partial differential equation [23]:

$$-k_t \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) = Q, \quad Q = P/V \quad (1)$$

where T denotes the temperature field, k_t represents the material thermal conductivity, and Q is the volumetric heat source density. Q is determined by the ratio of the heat dissipation power P to the the heat source volume V . (x, y, z) uses the Cartesian coordinate system.

In addition to internal heat conduction, ICs typically require consideration of heat exchange boundary conditions with the external environment [24]. Here, we employ the convective boundary condition to describe the heat flux between the chip and the environment medium, which is expressed as Eq. (2):

$$-\mathbf{n} \cdot (k_t \nabla T) = h(T_{\text{ext}} - T) \quad (2)$$

where $\mathbf{n}$ is the outward unit normal vector, k_t is the thermal conductivity, h is the convective heat transfer coefficient, and T_{ext} represents the external environment temperature.

In 2.5D ICs, due to the differences of material properties and geometries in chiplets, interposers, substrates, and the non-uniform heat source distribution, the heat conduction becomes a complex high-dimensional challenge in numerical analysis methods [25]. Traditional numerical FEM methods often require dense meshes to simultaneously capture local fine-grained gradients and global large-scale temperature distributions, leading to a significant increase in computational cost [26].

B. Point cloud

A point cloud is a discrete set of data points in a 3D coordinate system, represented as $\mathcal{P} = \{p_i \in \mathbb{R}^d \mid i = 1, \dots, L\}$, where each point p_i may include coordinates (x, y, z) and optional feature attributes (e.g., intensity, material properties) [27], d represents dim. In IC design, point clouds hold the potential for representing physical structures, layout, and signal transmission paths. Compared to traditional grid- and graph-based representations, point clouds offer advantages [28], [29]: 1) *Sparsity*: Efficiently represent complex 3D structures with minimal data, reducing storage and processing computational overhead critical for large-scale ICs. 2) *Flexibility*: Adapt to various structures of ICs and can model unstructured 3D structures (e.g., bumps, stacked dies) without explicit meshing. 3) *High Precision*: Accurately preserve fine geometric details, enhancing the precision and reliability in the modeling of ICs.

III. METHODOLOGY

The overall pipeline of the thermal prediction framework is illustrated in Fig. 1. This paper constructs point clouds through adaptive sampling for 2.5D ICs, ensuring the fidelity of multi-scale unstructured geometries. Furthermore, the point clouds are processed through Therm-PCT, which includes modules for heat diffusion via adaptive multipath-coupled diffusion (AMD), gradient recovery via wavelet-based fine-grained recovery (WFR), and heterogeneous adaptation via thermal-aware Mixture-of-Material-Experts (TA-MoME). They incorporate accomplish high-precision thermal and its gradient prediction for large-scale unstructured 2.5D ICs. Finally, the framework achieves temperature field generation for 2.5D ICs.

A. Unstructured point cloud construction

Point cloud sampling: For a given 2.5D IC design, a point cloud is constructed through sampling in its internal domain and boundaries. The design's multi-scale geometry, including micrometer-scale bumps and millimeter-scale substrates, is sampled using spacing that adapts to the geometric scales. Finer spacing is assigned to smaller-scale structures (e.g., bumps) to capture intricate details, while coarser spacing is assigned to larger-scale components (e.g., substrates) to reduce redundancy, thereby ensuring efficient sampling and geometric fidelity. Moreover, thermal related variables including thermal conductivity k_t , heat flux h , and heat source dissipation power P in the heat equations are retained as original input attributes

at each sampling point. It ensures that physical properties are directly encoded into the point cloud feature representation. Representing the 2.5D IC by an unstructured point cloud instead of the mesh reduces computational memory requirements.

1D Serialization encoding: The unordered nature of conventional point clouds results in the loss of spatial characteristics essential for heat propagation [30]. To address this issue, unordered point clouds are serialized into M ordered 1D sequences via multiple 3D space-filling curve strategies [28], including Hilbert and Z-order curves as well as axis-swapping techniques. Then the different spatial proximity of the original geometry can be preserved (e.g., Hilbert curves maintain local locality, Z-order curves capture long-range correlation), which is essential for capturing complex geometry and physical relationships. Thus, the 1D serialization encoding method can support spatially ordered sequence modeling for the unstructured geometry in 2.5D ICs.

B. Geometry-material aware point cloud transformer: Therm-PCT

In this section, the Therm-PCT architecture incorporates a hierarchical encoder-decoder design for thermal feature processing in 2.5D ICs across four stages, shown in Fig. 1. The encoder captures thermal patterns from coarse to fine, while the decoder gradually reconstructs complete thermal distributions. Each stage employs grid-based pooling/unpooling operations and point transformer blocks to learn multi-scale thermal patterns while maintaining spatial relationships. The point transformer block includes three modules to learn complicated thermal patterns: 1) **Thermal-Diffuse**, which uses adaptive multipath-coupled diffusion to learn interactions of multiple 1D heat conduction serialization pathways; 2) **Gradient-Recover**, which recovers high-frequency thermal gradient details via wavelet-based fine-grained recovery module; and 3) **Hetero-Adapt**, which learns heterogeneous thermal distributions via dynamically routing material-specific experts.

1) **Thermal-Diffuse: adaptive multipath-coupled diffusion (AMD):** After efficient representations of large-scale unstructured geometries, accurate thermal prediction for large-scale 2.5D ICs requires comprehensive feature extraction of spatially unstructured point clouds. After encoding input point cloud into $\mathbf{F}'_i$ through implicit conditional positional encoding (CPE) [31], the input $\mathbf{F}'_i$ is been serialized into multiple ordered 1D sequences. Each 1D serialization sequence represents a kind of hypothesized heat conduction pathway. However, 2.5D IC heat diffusion propagates through multiple spatial pathways influenced by chiplet layouts and material interfaces. In this case, one single 1D sequence token interaction in the attention operator results in a loss of local 3D topological relationships. It ignores multiple heat path coupling interactions for heat diffusion.

To address the challenge, we propose adaptive multipath-coupled diffusion (AMD), it achieves the fusion of attention diffusion features in multiple thermal pathways (Fig. 1), enabling scalable and topology-aware thermal feature extraction.

Adaptive serialization gate: After initializing multiple serialization pathways, a lightweight gate network assesses the relevance of serialization paths to the input geometry and material distribution. It can select top- K paths via the learnable gating parameter g_j at inference phrase. The selection allows the attention computation to focus on physically plausible thermal pathways, which reduces computation redundancy and preserves critical topological relationships. The gating compatibility score s_j in the top- K path at inference phrase is computed as Eq. (3):

$$s_j = \underset{j \in \{1, \dots, M\}}{\operatorname{argtopK}} (\operatorname{softmax}(g_j)) \quad (3)$$

where a learnable gating parameter vector $g \in \mathbb{R}^M$ is defined, and M is the total number of preset serialization paths. The top- K strategies with the top highest K scores are selected to construct K parallel attention branches.

Self-attention operator: To overcome $O(n^2)$ computational complexity limitations in transformers, especially when processing large-scale point clouds [32], a lightweight self-attention operator Flash Attention [33] is applied. The operator extracts the thermal diffusion features of the top- K pathways from parallel K branches. And it mitigates memory overhead through tiling and recomputation strategies. Then thermal diffusion feature outputs from all K branches are aggregated via learnable gating weights s_j . This attention design reduces memory overhead to the linear scale of the point number while adaptively combining salient thermal pathways. Specifically, the attention computation for the j -th branch is defined as Eq. (4).

$$\mathbf{F}_i^{\text{attn}} = \sum_{j=1}^K s_j^{\text{norm}} \cdot \bigcup_{m,n} \operatorname{softmax} \left(\frac{\mathbf{q}^{(m)} (\mathbf{k}^{(n)})^\top}{\sqrt{d}} \right) \mathbf{v}^{(n)} + \mathbf{F}_i' \quad (4)$$

where $\bigcup$ denotes block-wise computation via tiling strategy. The block-wise query $\mathbf{q}^{(m)}$, key $\mathbf{k}^{(n)}$, and value $\mathbf{v}^{(n)}$ are linearly projected from $\mathbf{F}_i$ and respectively partitioned into M, N blocks along the spatial dimension. Here, $M = N = \lceil L/B \rceil$, where L is the sequence length and B is the block size governing memory efficiency. The normalized gating weight s_j^{norm} is computed via $\frac{\exp(s_j)}{\sum_{k=1}^K \exp(s_k)}$ with gating compatibility scores s_j for weighting branches.

2) **Gradient-recover: wavelet-based fine-grained recovery (WFR) module:** Although self-attention mechanisms in thermal diffusion can capture long-range dependencies, their spectral

smoothing effect tends to attenuate local details, affecting the recovery of high-frequency gradients [34], [23]. Modern 2.5D IC thermal simulations require precise resolution of multi-scale thermal gradients, from micron-level hotspots to full-chip thermal profiles. Additionally, constrained by homogeneous receptive fields and fixed kernel assumptions [21], conventional deep learning architectures struggle to simultaneously perceive changes in the thermal scales from high-frequency thermal details (e.g., localized bump hotspots own rapid gradients) to low-frequency gradual thermal variations [15]. To address this, we propose the wavelet-based fine-grained recovery (WFR) module (Fig. 2). It explicitly decouples and refines fine-grained thermal features through high-frequency wavelet-domain enhancement, achieving the capability of fine-grained gradient awareness.

The WFR module integrates a dual-branch wavelet decomposition pipeline to process point cloud tokens $\mathbf{F}_i^{\text{attn}} \in \mathbb{R}^{L \times d}$ in Eq. (4) after Thermal Diffusion. WFR first splits the features into low- and high-frequency components via 1D Discrete Wavelet Transform (DWT) in Eq. (5). Specifically, for each token $\mathbf{F}_i^{\text{attn}}$, DWT decomposes its feature into trend coefficient features $\mathbf{F}_i^{\text{low}}$ encoding low-frequency thermal trends and detail coefficient features $\mathbf{F}_i^{\text{high}}$ capturing high-frequency gradients.

$$\mathbf{F}_i^{\text{high/low}}[n] = \sum_{t=-L/2}^{L/2} \mathbf{F}_i^{\text{attn}}[n-t] \mathcal{F}^{\text{high/low}}(t) \quad (5)$$

where $\mathbf{F}_i^{\text{high/low}}[n]$ represent the detail and trend coefficient features, respectively. $\mathcal{F}^{\text{high/low}}$ denote the high-pass and low-pass filters. L is the filter length, which determines the number of coefficients used in the convolution operation.

The high-frequency branch employs a coefficient-recover network $\tilde{\mathbf{F}}_i^{\text{high}} = \mathcal{M}(\mathbf{F}_i^{\text{high}})$ to enhance localized gradient interactions via nonlinear learning, where network $\mathcal{M}(\cdot)$ includes two linear and activation layers. And the low-frequency branch applies a linear transformation $\tilde{\mathbf{F}}_i^{\text{low}} = \mathcal{W}_r(\mathbf{F}_i^{\text{low}})$ to preserve the previous global thermal attention features.

Finally, the refined frequency coefficient features $\tilde{\mathbf{F}}_i^{\text{low}}, \tilde{\mathbf{F}}_i^{\text{high}}$ are fused via inverse wavelet transform (IDWT) and projected to the original spatial domain using a fusion layer $\mathcal{G}(\cdot)$ as Eq. (6):

$$\hat{\mathbf{F}}_i = \mathcal{G}(\text{IDWT}(\tilde{\mathbf{F}}_i^{\text{low}}, \tilde{\mathbf{F}}_i^{\text{high}})) + \mathbf{F}_i^{\text{attn}} \quad (6)$$

where $\mathcal{G}(\cdot)$ includes the two linear layers and a GELU nonlinear activation, and $\hat{\mathbf{F}}_i$ is the output of the fused feature.

3) **Hetero-Adapt: thermal-aware Mixture-of-Material-Experts (TA-MoME):** The heterogeneous integration of chiplets and materials in 2.5D/3D ICs introduces differential thermal gradients from different material conductivities. The high power density regions (e.g., compute units) exhibit steep temperature gradients, while low-activity areas (e.g., substrates) demonstrate gradual thermal variations. Traditional neural operators fail to adaptively resolve such gradient disparities due to parameter sharing and structural homogenization [21]. In this section, we design the TA-MoME adapter to adapt to different thermal gradients in different materials, as detailed in Fig. 3. This TA-MoME dynamically adapts to the

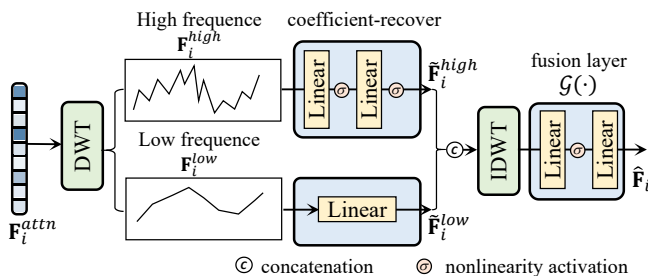


Fig. 2: Frequency branches of wavelet-based fine-grained recovery module.

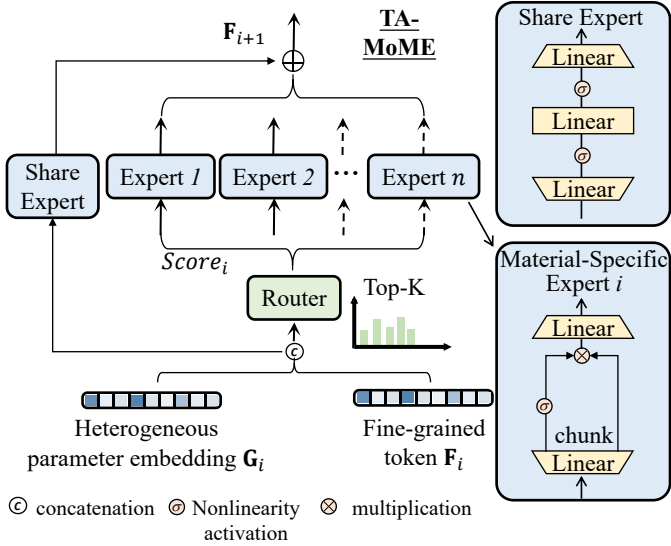


Fig. 3: Details of thermal-aware Mixture-of-Material-Experts (TA-MoME).

heterogeneous thermal properties from the material layout by leveraging the mixture-of-experts (MoE) framework. It routes point tokens to specialized experts, and each expert specializes in modeling the thermal characteristics of specific materials and regions.

The TA-MoME consists of a material-expert top-K router and multiple expert modules. And the experts include the shared expert and the material-specific expert.

Material-expert top-K router: The router assigns each token to the top K relevant experts based on a compatibility score Score_i derived from heterogeneous material embeddings $\mathbf{G}_i$ and local detailed token features $\hat{\mathbf{F}}_i$. It activates a subset of experts by Eq. (7). The top-K expert selection ensures adaptive specialization, which tokens from high-conductivity materials (e.g., copper) are routed to experts optimized for rapid heat dissipation, while low-conductivity thermal mediums activate the experts focused on interfacial heat accumulation.

$$\text{Score}_i = \text{softmax}(W_r \cdot (\mathbf{G}_i | \hat{\mathbf{F}}_i)) \quad (7)$$

where $\mathbf{G}_i$ encodes the material-specific thermal parameters (e.g., conductivity) into a learnable embedding space, while $\hat{\mathbf{F}}_i$ represents spatial-semantic features from preceding stages. W_r is the linear transformation. $|$ denotes concatenation.

Shared and Material-Specific Experts: TA-MoME employs a hybrid expert pool: (a) a shared expert that models universal heat transfer mechanisms. It utilizes a standard Feed-Forward Network (FFN) with two linear layers and GELU nonlinearity activation to capture universal thermal transport patterns, such as the universal laws of heat conduction across all materials. (b) n material-specific experts optimized for regions with thermal differences (e.g., copper interconnects requiring rapid heat dissipation or convective boundary). It utilizes a gated dual-branch module with sigmoid-activated linear projections and element-wise multiplication to enhance sensitivity to localized thermal gradients. The gating mechanism in the

module selectively emphasizes/suppresses features to model abrupt thermal shifts in high-power regions (e.g., hotspots near chiplets) or gradual variations in substrates. Finally, their outputs are aggregated as $\mathbf{F}_{i+1}$ via summation.

4) **Pooling and unpooling:** To further enable multi-scale thermal pattern learning while preserving critical geometric relationships, we integrate grid-based pooling/unpooling operations before/after the point transformer blocks in each stage, as shown in Fig. 1. The pooling operation reduces the spatial resolution of the point cloud, aggregating average local features into coarser representations while preserving critical topological information. This is achieved through a sparse voxelization process, where points within a voxel are embedded into a unified feature vector. The pooled features are then propagated through the transformer blocks, enabling hierarchical feature extraction at multiple scales. The unpooling operation reconstructs the feature space to its original resolution by copying features from coarser levels. This is implemented via a grid-based unpooling mechanism, which preserves the fine-grained details lost during pooling. Moreover, the unpooling features are combined with the same-resolution features from the encoder stages, achieving that both the encoder and decoder thermal characteristics are captured effectively.

5) **hybrid loss function:** To jointly optimize thermal field accuracy and expert load balancing in the MoME adapter, a hybrid loss function was employed, combining mean squared error (MSE) with an auxiliary expert load balancing loss. The hybrid loss is formulated as Eq. (8):

$$L_{\text{hyb}} = L_{\text{MSE}} + \lambda \cdot \sum_{e=1}^E f_e \log \frac{f_e}{1/E}, \quad (8)$$

where E is the number of experts, f_e denotes the frequency of expert e being selected within a batch, and λ is a balancing coefficient.

IV. EXPERIMENTS AND RESULTS

A. Experiments setting

A benchmark dataset of 8,500 2.5D IC configurations was generated using an automatic script with COMSOL Multiphysics software [10]. The dataset adopts a variety of randomly combined 2.5D design layouts. i). The geometry of the designs consists of unstructured components and layered structures: The components comprise silicon-based chiplets, microbump arrays, and C4 bump arrays (Cu material); while layered structural materials include a flame retardant 4 (FR4) substrate, a silicon interposer, and thermal interface materials (TIM). ii). For thermal source configuration, the designs in the dataset include 1 to 16 silicon heat source units, generated via Latin Hypercube Sampling (LHS) for uniform distribution in the dataset. The unit power follows a normal distribution ($\mu = 1\text{W}$, $\sigma = 0.3\text{W}^2$). Moreover, the dimensions of chiplet heat sources range from 1 mm to 6 mm, maintaining a 0.05 mm edge margin. iii). The air heat flux boundary conditions at the top/bottom of 2.5D ICs are randomized using LHS within the range of 300–600 W/m^2 .

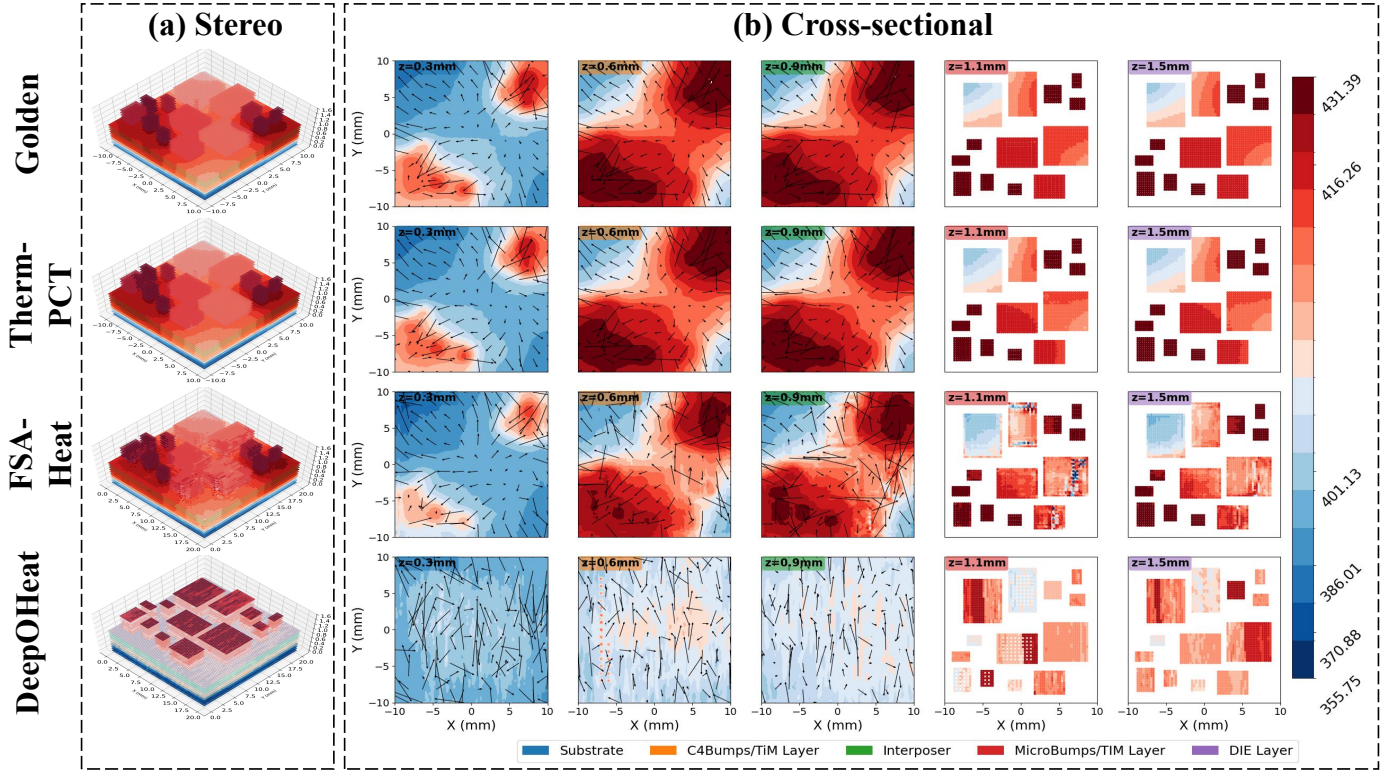


Fig. 4: Thermal and its gradient prediction comparisons between ours and others in a visualization of 2.5D IC with 13 heat sources. (a) and (b) respectively show the stereo visualization and cross-sectional slices of the predicted thermal fields by different methods. In (b), the slices consist of the top slice of each material (z-dimension). The legends in different colors correspond to different materials.

TABLE I
THERM-PCT MODULE CONFIGURATION AND HYPERPARAMETERS.

Module	Parameter	Value
Embedding	Input/Output Channels	3/32
	Block Numbers	Enc (2,3,3,2) Dec (2,2,2,2)
Encoder/Decoder	Output Channels	Enc (64,128,256,512) Dec (256,128,64,32)
		Attention Heads
	Patch Size/Ratio	Enc (4,8,16,16) Dec (16,8,4,4)
		1024/4
TA-MoME	Special Experts	nums 4 ratio 2
		General Experts
	ratio	nums 1 ratio 4
		1
WFR	Channel Vary Ratio	1
Prediction Head	Input/Output Channels	32/1

The model was trained for 300 epochs in 40 hours on two NVIDIA RTX 3090 GPUs using AdamW ($lr = 0.002$) with OneCycleLR scheduling and mixed-precision training (FP16). Gradient clipping was applied to ensure stable convergence during training.

The model employs an encoder-decoder transformer architecture. The detailed configuration and hyperparameters are as Table I, where the ratio denotes the ratio of the dimension expansion of the intermediate hidden layer of the module.

TABLE II
ACCURACY COMPARISON EXPERIMENT.

Metric	Method				
	Ours	FSA-Heat	DeepOHeat	GCN+PNA	COMSOL
MAE ↓	0.2319	1.0553	22.9356	OOM	GT
RMSE ↓	0.3822	2.3865	24.4706	OOM	GT
MAPE ↓	0.0931%	0.2874%	6.3655%	OOM	GT
MaxD ↓	0.3281	1.5092	22.8671	OOM	GT

And embedding consists of a sparse convolution layer and 2 transformer blocks with 2 attention heads.

B. Accuracy and generalization comparisons

To comprehensively evaluate Therm-PCT's performance in 2.5D IC thermal analysis, we conducted extensive experiments to compare the accuracy and generalization capabilities against ground-truth (GT) COMSOL and newly-proposed deep learning-based methods, including DeepOHeat, FSA-Heat, and GCN+PNA [11], [15], [16]. All deep learning methods used the same dataset. Note that, to train the structured CNN-based FSA-Heat on unstructured geometries, equally-sized input image slices are uniformly filled by point clouds.

Accuracy comparison and analysis: We evaluate Therm-PCT against all other baselines using four metrics—Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Maximum Temperature Deviation (MaxD), and Mean Absolute Percentage Error (MAPE). MAE measures average error

magnitude, RMSE highlights extreme errors, MaxD captures the top 5% worst-case errors for hotspot analysis, and MAPE evaluates relative error.

As illustrated in Fig. 4 and Table II, Therm-PCT achieves significant improvements in thermal prediction accuracy against other DL baselines. From the stereo visualization (Fig. 4(a)), it can be seen that our method exhibits continuity and fidelity in predicting both transverse and longitudinal temperature distributions and temperature gradients, especially in the top 2.5D IC where die heat sources are concentrated. In contrast, the structured FSA-Heat method's temperature predictions show point-wise fragmentation in the thermal distribution, indicating its inadequate performance in predicting temperature gradients in the unstructured 2.5D ICs. Cross-sectional slices (Fig. 4(b)) show more detailed local thermal fields in different materials. Our method outperforms other DL approaches in better details of temperature and gradient within the hotspot regions of various materials, particularly in resolving sharp thermal gradients at material interfaces (e.g., MicroBumps/TIM, interposer). Moreover, DeepOHeat adopts the MLP-operator architecture for physics-unsupervised training to handle simple experimental ICs in [11], which include fixed thermal conductivity and the cubic structure. However, the distortion of DeepOHeat's prediction slices indicates that the simple architecture and physics-unsupervised training might not be powerful enough to handle such complex IC settings with more complex geometries, thermal parameters, and higher resolution.

Quantitative results in Table II highlight Therm-PCT's comprehensive performance across all metrics: it reduces the MAE, RMSE, MAPE, and MaxD by 78.03%, 84.00%, 67.61%, and 78.25% compared to FSA-Heat, which indicates its critical capability to localize hotspots in ICs. And GCN+PNA encountered out-of-memory (OOM) issues, showing the unsatisfactory performance of graph-based methods for large-scale point clouds. Additionally, the speed of Thermal-PCT is 0.38s compared to the 56s of the GT software COMSOL, achieving a 147 $\times$ acceleration ratio.

Generalization testing on larger resolutions without retraining: In early chip design, engineers often iterate with geometric models of varying precisions from rough layouts to refined structures. To validate the generalization of Therm-PCT at more refined geometric resolutions, we performed generalization experiments by adjusting the point cloud sampling density without retraining. For each 2.5D IC configuration in our benchmark dataset, we generated three resolution levels: standard (original sampling, up to 86,395 points), high (about 3.0 $\times$ dense sampling, up to 258,316 points), and ultra (about 5.7 $\times$ dense sampling, up to 496,365 points).

As shown in Table III, as the point cloud sampling density increases from 80 K-scale standard resolution to 0.4 M-scale ultra resolution, the error metrics (RMSE, MAE, MAPE) experience slight increases, but performance remains acceptable. This suggests that Therm-PCT has the potential to adapt to geometric in larger resolutions, maintaining relatively accurate predictions even with larger-scale point clouds. Additionally,

TABLE III
RESOLUTION GENERALIZATION PERFORMANCE COMPARISON.

Resolution		Standard (80K scale)	High (0.25M scale)	Ultra (0.4M scale)
Metrics	MAE $\downarrow$	0.2319	0.5796	0.5358
	RMSE $\downarrow$	0.3822	0.6456	0.6187
	MAPE $\downarrow$	0.0931%	0.1594%	0.1518%
	MaxD $\downarrow$	0.3281	0.5257	0.5057
FSA-Heat	MAE $\downarrow$	1.0553	1.3031	1.8142
	RMSE $\downarrow$	2.3865	1.4983	2.0924
	MAPE $\downarrow$	0.2874%	0.3558%	0.4953%
	MaxD $\downarrow$	1.5092	1.6421	2.3300

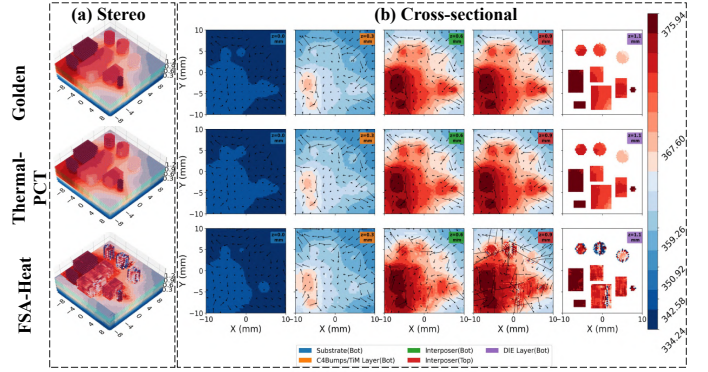


Fig. 5: The experimental results of thermal distribution and temperature gradient prediction under shape changes for Therm-PCT.

the variations in MaxD across resolutions indicate that Therm-PCT effectively identifies temperature field hotspots.

Generalization testing of different shapes without retraining: To validate Therm-PCT's robustness across diverse geometric configurations without retraining, we conducted shape-variation generalization experiments by testing on 1,200 unseen 2.5D IC layout datasets, where chiplet heat sources comprised 50% cylindrical and 50% rectangular cuboid geometries.

From Fig. 5, the proposed Therm-PCT still showed high thermal and gradient prediction accuracy in different material layers, even when the heat source shape undergoes significant changes. Notably, compared with the FSA-Heat, the model achieves precise predictions of cylindrical heat sources' thermal diffusion, capturing their symmetric temperature distribution and gradient patterns with small errors. This performance highlights Therm-PCT's potential to generalize effectively to unseen shapes while maintaining fidelity to various geometries' underlying thermal diffusion patterns.

C. Ablation Experiments

In our ablation study analysis of the Therm-PCT, Table IV reveals distinct module functions through component-wise removal. Removing the AMD module caused the comprehensive performance degradation. It causes 30.82%, 9.21%, 7.20%, 4.39% increase in MAE, RMSE, MAPE, and MaxD, indicating its significant foundational function in maintaining thermal propagation pathways' integrity. Disabling the WFR module led to a significant increase in MAE, RMSE, and MAPE (36.40%, 11.96%, 11.71%), while simultaneously worsening MaxD. It suggests that WFR reduces the overall error while

TABLE IV
ABLATION STUDY OF THERM-PCT CORE MODULES.

Variants	↓ MAE	↓ RMSE	↓ MAPE	↓ MaxD
Full Therm-PCT	0.2319	0.3822	0.0931%	0.3281
w/o AMD	0.3034	0.4174	0.0998%	0.3424
w/o WFR	0.3163	0.4279	0.1040%	0.2775
w/o TA-MoME	0.2744	0.3997	0.0962%	0.3438

recovering the high-frequency gradients, but may introduce some high-frequency noise. Eliminating TA-MoME resulted in an increase of all four metrics, highlighting its material-aware compensation mechanism. Next, further validation explanations regarding the respective thermal challenges addressed by the WFR and TA-MoME modules are as follows.

Gradient validation of WFR: To validate the WFR module's effectiveness in recovering high-frequency thermal gradients, we visually compared predictions on the C4 bumps and their TIM. The experiment evaluated three configurations: the golden, Therm-PCT, and Therm-PCT without the WFR module.

As shown in the zoomed-in hotspot of C4 TIM layer (Fig. 6(b)), the micro bump arrays display significant thermal gradients due to localized heat accumulation and conductivity mismatches. The zoomed-in micro bump TIM case reveals that the Therm-PCT without WFR displays blurred thermal contours and diminished high-frequency details, especially near micro bumps' edges and interfaces. In contrast, the full Therm-PCT's fine-grained thermal gradients closely match the ground truth. This improvement results from the WFR's dual-branch wavelet decomposition. By enhancing high-frequency subbands through coefficient recovery and inverse wavelet fusion, WFR mitigates spectral smoothing artifacts from conventional attention.

The results confirm that the WFR module significantly improves the network's capability to resolve micron-scale thermal discontinuities, which are critical for accurate hotspot localization in 2.5D IC designs.

Expert validation of TA-MoME: To verify the effectiveness of TA-MoME in adapting to heterogeneous thermal material, we analyze its expert routing behavior in the last transformer block of the final decoder stage (Fig. 7). The observed expert selection patterns include:

1) *High-Conductivity Regions Favor Expert2:* The bottom substrate (convective boundary) and structures requiring rapid

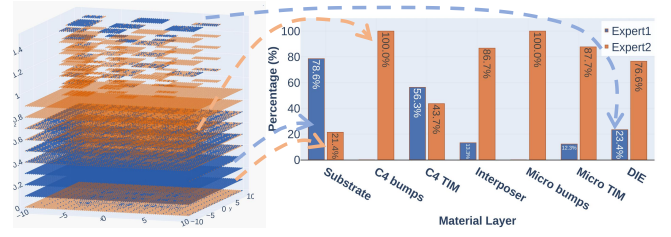


Fig. 7: A validation case of specialized expert router behavior in heterogeneous material layers in TA-MoME.

heat dissipation, such as C4 bumps and interposer, strongly prefer expert2. This matches the need to capture directional heat flux and sharp thermal gradients near active chiplets.

2) *Low-Conductivity Regions Activate Expert1:* Expert1 is predominantly activated in regions with low thermal conductivity or gradual thermal propagation, including the majority of the substrate. These layers involve dielectric materials where heat retention and smooth thermal gradients dominate.

3) *Hybrid Allocation for Interfacial Materials:* The C4 TIM reflects balanced routing, serving dual thermal roles: balancing between expert2-driven heat spreading in heterogeneous materials and expert1-driven interfacial heat accumulation.

These patterns validate TA-MoME's material-aware specialization: expert allocation aligns with material-driven thermal priorities, resolving limitations of homogeneous networks in 2.5D IC thermal prediction.

V. CONCLUSIONS

This paper presents Therm-PCT for large-scale unstructured 2.5D IC thermal analysis. To characterize large-scale unstructured complex geometries, we present the multiscale point-cloud representations. To learn adaptive thermal diffusion path interactions, adapt to differential thermal distributions among heterogeneous materials, and recover high-frequency fine-grained thermal gradients, Therm-PCT integrates three modules: the adaptive multipath-coupled diffusion (AMD) module, the thermal-aware Mixture-of-Material-Experts (TA-MoME) adapter, and the wavelet-based fine-grained recovery (WFR) module, which target respective thermal challenges. Experiments demonstrate that Therm-PCT achieves a 147× speedup over commercial software COMSOL while outperforming newly-proposed DL method FSA-Heat, with reductions in MAE and MaxD by 78.03%, 78.25%, respectively. Notably, it excels in recovering fine-grained thermal gradients and adapting to large temperature ranges due to heterogeneous materials. Furthermore, Therm-PCT generalizes robustly across varying geometric sample resolutions and unseen geometric variations. We hope Therm-PCT can contribute to accelerating thermal-aware optimization in early-stage 2.5D IC design workflows.

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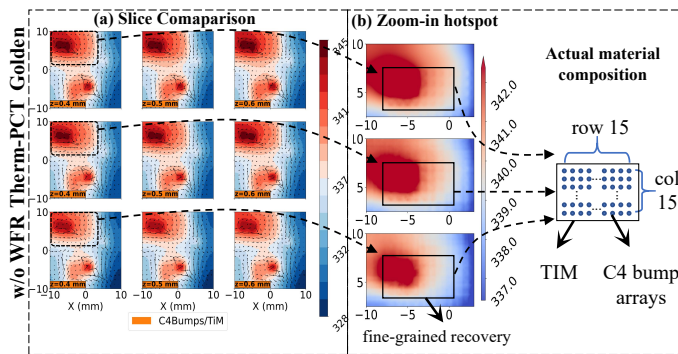


Fig. 6: A C4 bump/TIM layer thermal visualization (a) and its zoomed-in hotspot (b) to validate high-frequency gradient recovery with/without WFR.

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